

```
AppShellUI
+UStateController : UStateController
+selectedRegime : RegimeType
+tradingStatus : TradingStatus
+startInterfac() void
+hasActiveTradingProgress : RegimeType void
+startConfirmed() void
+stopConfirmed() void
```

```
RecentOrderUI
+UStateController : UStateController
+recentTrades : List<Trade> = []
+showAllTradingDetails() void
```

```
TradingHistoryUI
+Controller : UStateController
+selectedTimePeriod : HistoryPeriod = TODAY
+selectedSide : TradeSide = ALL
+displayedTrades : List<Trade> = []
+details : TradeDetailsResult = null
+selectedFilterPeriod : HistoryPeriod, side : TradeSide void
+displayTradingDetails() : TradeDetails Result void
+displayFilteredTradingDetails : List<Trade> void
+displayCSVExport() void
```

```
PopUI
+options : CSVExportOptions
+isVisible : bool = false
+validatedInterim : String = null
+exportStatus : CSVExportStatus = IDLE
+validateInterim() void
+changeCSVExportPeriod : CSVPeriod, startDate : LocalDate, endDate : LocalDate, fileName : String void
+exportConfirmed() void
+hasCSVExportOptions() : CSVExportOptions void
+displayAvailableExportPath : Path void
+refreshOptions : CSVExportOptions void
+showValidatedInterim() : ValidationResult void
+showExportProgress() void
+showExportReason : String void
+showExportCompletePath : Path void
```

```
UStateController
+USTM : USTM
+appShellUI : AppShellUI
+marketDataController : MarketDataController
+tradingController : TradingController
+tradingHistoryController : TradingHistoryController
+recentOrders : RecentOrdersUI
+tradingStatus : TradingStatus
+csvExport : CSVExport
+csvExportOptions : CSVExportOptions
+hasActiveTradingProgress : RegimeType void
+startTrading() void
+showAllTradingDetails() void
+hasActiveTradingProgress : HistoryPeriod, side : TradeSide void
+exportCSV() void
+hasActiveTradingProgress : HistoryPeriod, side : TradeSide void
+displayTradingDetails() : TradeDetails Result void
+displayFilteredTradingDetails : List<Trade> void
+exportCSV() void
```

```
TradingController
+TradingSTM : TradingSTM
+context : TradingContext
+account : Account
+position : Position
+marketSnapshot : MarketSnapshot
+apiGateway : APGateway
+tradingHistoryController : TradingHistoryController
+selectedRegime : RegimeType = null
+hasActiveTradingProgress : String = "TFT", Account
+selectedTradingProgress : RegimeType : TradingSTM
+startTrading() void
+stopTrading() void
```

```
MarketDataController
+apiGateway : APGateway
+webSocketGateway : WebSocketGateway
+marketSnapshot : MarketSnapshot
+regimeController : RegimeController
+symbol : String = "ETHUSD"
+interval : Set<Interval>
+initializeMarketData() : Symbol = "ETHUSD", MarketSnapshot
+execute() void
```

```
APGateway
+restClient : BinanceRESTClient
+symbol : String = "ETHUSD"
+loadAllKlineSymbol : String, limit : int, Map<Interval, List<Kline>>
+hasActiveTradingProgress : String = "TFT", AccountSnapshot
+getActiveTradingProgress : String, quantity : Decimal, OrderResult
+submitOrder() : Order, OrderResult
+getOrderResult() : String, orderID : Long, OrderResult
```

```
WebSocketGateway
+websocketClient : BinanceWebSocketClient
+buffer : Map<Interval, List<Kline>> = []
+lineSubscription : Subscription = null
+accountSubscription : Subscription = null
+startAllKlineBuffer() : String, intervals : Set<Interval>
+startAccountSubscription() : Subscription
```

```
RegimeController
+marketSnapshot : MarketSnapshot
+indicatorSnapshot : IndicatorSnapshot
+regimeSTM : RegimeSTM
+tradingController : TradingController
+recommendedRegime : RegimeType = null
+selectedRegime : RegimeType = null
+calculateRecommendation : MarketSnapshot : IndicatorSnapshot
+recommendedRegimeIndicators : IndicatorSnapshot : RegimeType
+lastRegimeType : RegimeType void
```

```
TradingHistoryController
+tradingHistory : TradingHistory
+performance : Performance
+account : Account
+repository : TradingHistoryRepository
+csvFileGateway : CSVFileGateway
+hasActiveTradingProgress : TradingHistory
+recentOrderExecution : Order, summary : ExecutionSummary, costBasis : Decimal void
+getTradingDetails() : HistoryPeriod = Today, side : TradeSide = ALL, TradeDetailsResult
+exportCSV() : CSVExportOptions, CSVExportOptions, CSVExportResult
```

```
TradingHistoryRepository
+storagePath : Path
+filePath : Path
+getTradingHistory() : List<Trade>
+saveTradingHistory() : long, trade : Trade void
+streamTradingQuery : TradingHistoryQuery : Stream<Trade>
```

```
CSVFileGateway
+filesystem : FileSystem
+chooseDirectory : Path
+writeCSVTrades : Stream<Trade>, options : CSVExportOptions : CSVExportResult
```

```
USTM
+currentDate : UDate
+runInitialEvent : UEvent, UTransactionResult
+handleEvent : UEvent : UTransactionResult
```

```
TradingSTM
+currentDate : TradingDate
+currentType : RegimeType
+getTradingStatus : RegimeType : TradingSTM
+runInitialEvent : TradingContext : TradingTMResult
+handleEvent : TradingContext : TradingTMResult
+handleEvent : TradingContext : TradingTMResult
+handleEvent : TradingContext : TradingTMResult
```

```
RegimeSTM
+currentDate : RegimeType
+runInitialEvent : IndicatorSnapshot : RegimeType
```

```
TradingContext
+account : Account
+position : Position
+selectedRegime : RegimeType
+selectedTimePeriod : HistoryPeriod = TODAY
+selectedSide : TradeSide = ALL
+displayedTrades : List<Trade> = []
+details : TradeDetailsResult = null
+selectedFilterPeriod : HistoryPeriod, side : TradeSide void
+displayTradingDetails() : TradeDetails Result void
+displayFilteredTradingDetails : List<Trade> void
+exportCSV() void
```

```
MarketSnapshot
+symbol : String = "ETHUSD"
+interval : Map<Interval, List<Kline>> = []
+currentTime : Decimal
+update() void
+updateKline : Map<Interval, List<Kline>>
+updateCurrentTime : Decimal
```

```
IndicatorSnapshot
+timeframe : Interval = FOUR_HOURS
+emaSeries : List<Decimal> = []
+emaSeries : Decimal
+emaSeries : Decimal
+emaSeries : Decimal
+update() void
+updateSeries : List<Decimal>
```

```
Order
+symbol : String
+side : OrderSide
+strategy : StrategyType
+requestedQuantity : Decimal
+marketPriceDecision : Decimal
+exchangeOrderID : Long = null
+status : OrderStatus = null
+filledQuantity : Decimal = 0
+filledAmount : Decimal = 0
+fillPrice : Decimal = null
+processOrder : Instant = null
+fill : List<Fill> = []
+fillReason : String = null
+orderSymbol : String, side : OrderSide, strategy : StrategyType, requestedQuantity : Decimal, marketPriceDecision : Decimal : Order
+updateOrderResult() : OrderResult void
+updateOrderResult() : OrderResult void
+updateOrderResult() : OrderResult void
+updateOrderResult() : OrderResult void
```

```
Position
+symbol : String = "ETHUSD"
+owner : PositionOwner = null
+quantity : Decimal = 0
+averageEntryPrice : Decimal = 0
+costBasis : Decimal = 0
+enteredAt : Instant = null
+status : PositionStatus = Closed
+exitReason : ExitReason = null
+updatePosition() : Decimal, Decimal
+updatePosition() : Decimal, Decimal
```

```
Trade
+orderID : Long
+executedAt : Instant
+side : OrderSide
+price : Decimal
+quantity : Decimal
+amount : Decimal
+fee : Decimal
+strategy : StrategyType
+marketPriceDecision : Decimal
+realizedPrice : Decimal = null
+realizedRate : Decimal = null
+exitReason : ExitReason = null
+TradeOrder : Order, summary : ExecutionSummary, realizedResult : RealizedResult : Trade
+update() void
```

```
Performance
+dailyReturnRate : Decimal = 0
+cumulativeReturnRate : Decimal = 0
+realizedPrice : Decimal = 0
+dailyFee : Decimal = 0
+averageEntryPrice : Decimal = 0
+averageExitPrice : Decimal = 0
+winningExitCount : int = 0
+losingExitCount : int = 0
+PerformanceTrades : List<Trade>
+calculatePerformanceSummary : ExecutionSummary, costBasis : Decimal, RealizedResult
+updatePerformance() : Performance
```

```
Account
+balances : Map<String, Decimal> = []
+currentPrice : Decimal = 0
+valuation : Decimal = 0
+update() void
+getHoldingQuantity : String = "TFT", Decimal
```

```
TradingHistoryQuery
+startDate : LocalDate
+endDate : LocalDate
+side : TradeSide
+TradingHistoryQuery(startDate : LocalDate, endDate : LocalDate, side : TradeSide = ALL) : TradingHistoryQuery
```

```
CSVExportOptions
+destination : Path = null
+period : CSVPeriod = Today
+startDate : LocalDate
+endDate : LocalDate
+fileName : String
+CSVExportOptions(startDate : CSVPeriod = Today, startDate : LocalDate = Today, endDate : LocalDate = Today, fileName : String = defaultFileName) : CSVExportOptions
+exportCSV() : CSVExportOptions, CSVExportOptions, CSVExportResult
+exportCSV() : CSVExportOptions, CSVExportOptions, CSVExportResult
+exportCSV() : CSVExportOptions, CSVExportOptions, CSVExportResult
```

```
TradingEvent
+eventID : int
+timestamp : DateTime
+sequenceNumber : int
+priority : ExecutionPriority
+eventID : String
+timestamp : DateTime
+sequenceNumber : int
```

```
TradingHistory
+tradingHistory : List<Trade>
+tradingHistory : List<Trade>
+tradingHistory : List<Trade>
+tradingHistory : List<Trade>
```